

Math 113 – Abstract Algebra

Homework 2 — my solutions

Fall 2026 ■ Keshav Ramamurthy

All assignments must be **turned in on Gradescope** by the deadline. Late assignments are not accepted.

Have questions? Work with your peers, attend office hours, check out the [Student Learning Center](#), and ask in discussion section!

Problems for credit. Turn in these problems; some subset of them will be graded.

1. Fraleigh Exercise 4.6
2. Fraleigh Exercise 4.10
3. Fraleigh Exercise 4.18
4. Fraleigh Exercise 4.32
5. Fraleigh Exercise 5.2
6. Fraleigh Exercise 5.8
7. Fraleigh Exercise 5.16
8. Fraleigh Exercise 5.50

Practice problems. The following problems will not be graded, and should not be turned in, but are recommended as good practice problems. Solutions to most odd-numbered exercises are available in the back of the textbook.

- Fraleigh Exercises 4.1, 4.3, 4.7, 4.11, 4.13, 4.15, 4.19, 4.20, 4.23, 4.29, 4.31, 4.33
- Fraleigh Exercises 5.1, 5.5, 5.9, 5.13, 5.15, 5.19, 5.27, 5.39, 5.41, 5.53, 5.57

Solutions

Problem 1 (Fraleigh Exercise 4.6)

Solution. **a)** The three necessarily axioms are associativity, existence of an inverse, and an identity element (and closure under the operator).

We try this given the operator $*$ defined on $\mathbb{C}$ such that $a * b = |ab|$

We have

$$a * (b * c) = a * (|bc|) = |a|bc| = |abc|$$

and

$$(a * b) * c = (|ab|) * c = ||ab|c| = |abc|$$

so associativity holds.

We then see if there exists an element $e * a = a$ for all $a \in \mathbb{C}$ and we see that $\forall k \in \mathbb{C}, |k| \in \mathbb{R}$.

Thus, we can easily formulate a counterexample with a nonzero complex part, such as $1 + i$, and we see that solving $|e \cdot (1 + i)| = 1 + i$ which has no solutions for $e \in \mathbb{C}$. $\square$

Problem 2 (Fraleigh Exercise 4.10)

Solution. We first check that addition is closed under $n\mathbb{Z}$.

For $a, b \in n\mathbb{Z}$,

We have $\exists a', b' \in \mathbb{Z}$ such that $a'n = a, b'n = b$. by definition of $n\mathbb{Z}$. Then, for $a, b \in n\mathbb{Z}$, so

$$a + b = na' + nb' = n(a' + b') \text{ and since } a' + b' \in \mathbb{Z} \implies n(a' + b') \in n\mathbb{Z}$$

0 is the identity element because $\forall k \in n\mathbb{Z}, 0 + k = k$

Associativity also pretty easily holds because of associativity on $(\mathbb{Z}, +)$, as

$$a, b, c \in \mathbb{Z} \implies na, nb, nc \in n\mathbb{Z} \text{ and } na + (nb + nc) = (na + nb) + nc = na + nb + nc$$

Notice also the existence of an additive inverse, as

$$a \in \mathbb{Z} \implies -a \in \mathbb{Z} \implies n(a), -n(a) \in n\mathbb{Z} \text{ and } -n(a) + na = 0$$

Thus, all the axioms hold, and $(n\mathbb{Z}, +)$ is a group.

b) We subsequently will show that $(n\mathbb{Z}, +) \simeq (\mathbb{Z}, +)$. To do so, we have to prove injectivity, surjectivity, and homomorphism. For the homomorphism condition, we have to show for $(\mathbb{Z}, +)$ and $(n\mathbb{Z}, +)$,

$$f(x * y) = f'(x) *' f'(y)$$

or in this case:

$$f(x + y) = f(x) + f(y) \rightarrow n(x + y) = nx + ny$$

Claim. The function $f : \mathbb{Z} \rightarrow n\mathbb{Z}$ is bijective.

Proof. Claim. injectivity: For $x, y \in \mathbb{Z}$ where $f(x) = f(y) \implies x = y$

Proof. We have since $n > 0$ that

$$nx = ny \implies n(x - y) = 0 \implies x = y$$

□

Claim. Surjectivity: For any

$$a \in n\mathbb{Z}, \exists k \text{ such that } nk = a$$

Proof. This follows directly from the definition of the set $n\mathbb{Z} = \{ni : i \in \mathbb{Z}\}$ as n must divide every element in $n\mathbb{Z}$ meaning $k = \frac{a}{n} \in \mathbb{Z}$ and $f(k) = a$

□

□

Thus we are done.

□

Problem 3 (Fraleigh Exercise 4.18)

Solution. We wish to see if all $n \times n$ matrices with determinant 1 or -1 under matrix multiplication is a group.

Any matrix satisfying these conditions has an inverse as it has a nonzero determinant, and this inverse for a matrix A has the condition $\det(A^{-1}) = \frac{1}{\det A}$ which is either 1 or -1, closed under this condition.

Along with this, matrix multiplication holds under associativity and notice the identity matrix

$$I_n = \begin{pmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{pmatrix}$$

satisfies the determinant condition (has determinant 1) and for every matrix E that satisfies the conditions $I \cdot E = E$.

We establish closure: For matrices A, B satisfying these conditions, notice $\det(AB) = \det(A)\det(B)$ and since the product of 2 numbers in $\{-1, 1\}$ is in $\{-1, 1\}$, so AB satisfies the original conditions (as the product of these is also an n by n matrix)

Thus, we claim this is a group.

□

Problem 4 (Fraleigh Exercise 4.32)

Solution. We are given that G is a group, and that $\forall x \in G, x * x = e$. We first consider manipulating $(a * b) * (a * b) = e$ because by closure on G ,

$$\exists w \in G, a * b = w \rightarrow w * w = e.$$

Manipulating that expression yields

$$(a * (b * a) * b) \rightarrow e \rightarrow a^{-1} * (a * (b * a) * b) * b^{-1} = a^{-1} e b^{-1}$$

or

$$(b * a) = a^{-1} * e * b^{-1} = a^{-1} * b^{-1}$$

But we see that $a * a = e \implies a = a^{-1}$ due to the uniqueness of the inverse. So we have that $(b * a) = a * b$, proving commutativity and making it an Abelian Group. $\square$

Problem 5 (Fraleigh Exercise 5.2)

Solution. We wish to show if $\mathbb{Q}^+$ is a subgroup of $\mathbb{C}$ under $+$.

This is not true because the identity 0 in $\mathbb{C}$ and the subsequent existence of the additive inverse in $\mathbb{C}$, the additive inverse of $\mathbb{Q}^+$ does not exist, as negative numbers are not in $\mathbb{Q}^+$ (and the identity $0 \notin \mathbb{Q}^+$) $\square$

Problem 6 (Fraleigh Exercise 5.8)

Solution. $GL(n, \mathbb{R})$ is the group of $n \times n$ matrices with nonzero determinant.

We wish to show if (we'll denote H), the set of $n \times n$ matrices with determinant 2 is a subgroup of G .

This is immediately false, because we have that for matrix $M \in H$, $\det M^{-1} = \frac{1}{\det M} = \frac{1}{2}$ which is not in H , so H is not closed under inverses. (an example being:)

$$M = \begin{pmatrix} 2 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{pmatrix}$$

$\square$

Problem 7 (Fraleigh Exercise 5.16)

Solution. a) We consider the subset denoted H , $H = \{f \in \hat{F} \text{ such that } f(1) = 1\}$.

We claim it is not a subgroup of F under addition.

The identity function in F would be $\forall w \in \mathbb{R}, f(w) = 0$.

This identity does not exist in H as $\hat{F}$ describes functions that have a nonzero value at every point in $\mathbb{R}$, thus no function in H can be an identity. $\square$

Problem 8 (Fraleigh Exercise 5.50)

Solution. If the subset is closed under the binary operator $*$ of G , we have that $\forall a, b \in H, a * b \in H$. We consider an arbitrary element $a \in H$. This will be the key to our argument, as we will prove for an arbitrary element, it is closed under H , and has an identity and inverse.

Notice that H is finite. We can then repeatedly exponentiate a .

$$a, a^2, a^3, a^4, \dots$$

and notice this must have two powers that are the same by the pigeonhole principle.

We have $\exists x, y \in \mathbb{Z}^+$ such that $a^x = a^y$ and WLOG assume $x < y$. Then, notice $a^{y-x} = e$, so H has an identity.

We use the variable $m = y - x$.

If m is 1, notice that $a = e$, and thus a is also its own inverse,

If m is not 1, notice that $a * a^{m-1} = a^{m-1} * a = a^m = e$, so a^{m-1} is the inverse of a and recall it is closed under exponentiation. $\square$